Homework 3 Code

Matthew Schoen

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Part 1: Set Up

```
# Reading in necessary packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)

# Storing kelp data as specified object
kelp <- read_csv(
  here("data", "temp-kelp.csv"))

# Storing personal data as specificed object
step_count <- read_csv(
  here("data", "MyData_HW3_Format.csv"))
```

Part 2: Problems

Problem 1 - Giant Kelp Fronds

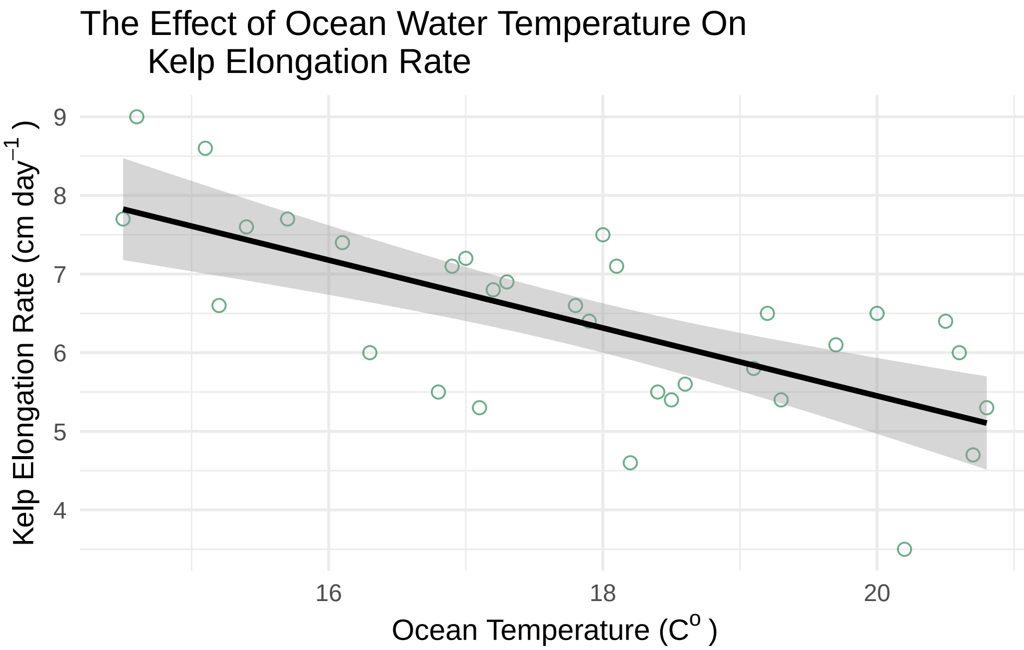
a. An Appropriate Test

To evaluate the strength of the relationship between temperature (C^o) and giant kelp frond elongation rate ($cmday^-1$), a Pearson correlation test and a Spearman rank correlation test would both be suited to use. The Pearson correlation test will evaluate whether there is a linear relationship between the two continuous variables while also assuming a normal distribution and linear relationship and the Spearman-rank correlation test will evaluate if a relationship exists between the two variables without requiring a normal distribution. As both ocean temperature and elongation rate are continuous and quantitative variables, both tests could be used depending on whether the data is normally distributed and linear.

b. Create a Visualization

```r
# Creating a ggplot base layer using kelp dataset
  ggplot(data = kelp,
         mapping = aes(x = temp_c,
                       y = kelp_elong)) +

  # First layer is a scatter plot to visualize individual observations
  geom_point(size = 2,
             alpha = 0.7,
             color = "#2E8B57",
             shape = 21) +
  # Second layer is a line of best fit using a linear regression
  geom_smooth(method = "lm",
              color = "black",
              linewidth = 1
             ) +
  # Using custom theme
  theme_minimal() +
  # Relabeling x & y axes
  labs(x = expression("Ocean Temperature (C"^o~")"),
       y = expression("Kelp Elongation Rate (cm day"^-1~")"),
       title = "The Effect of Ocean Water Temperature On
       Kelp Elongation Rate")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

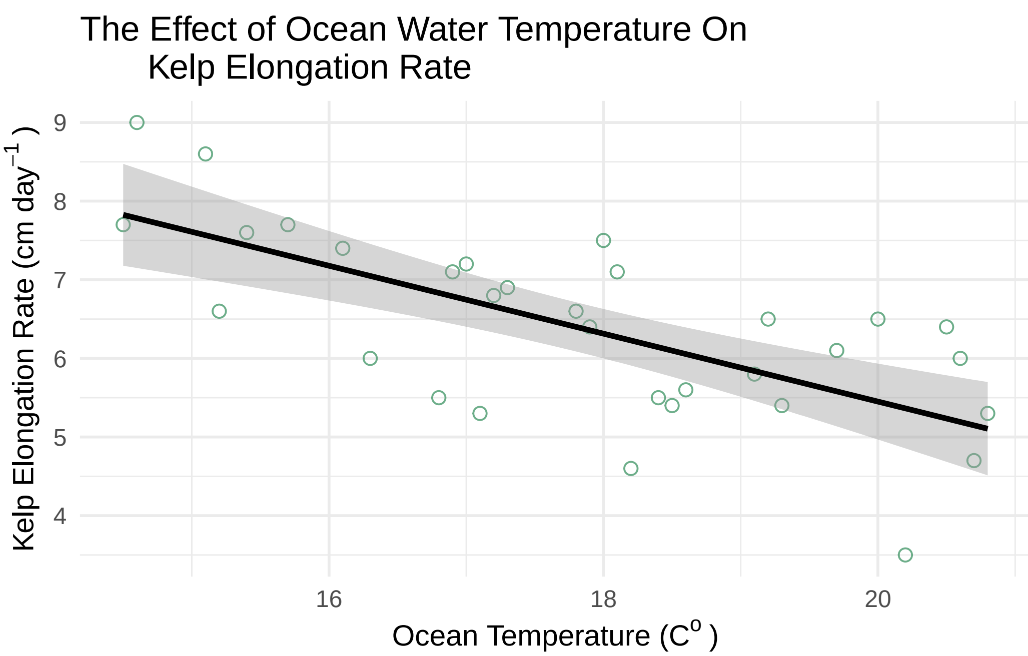


c. Check Your Assumptions and Run Your Test

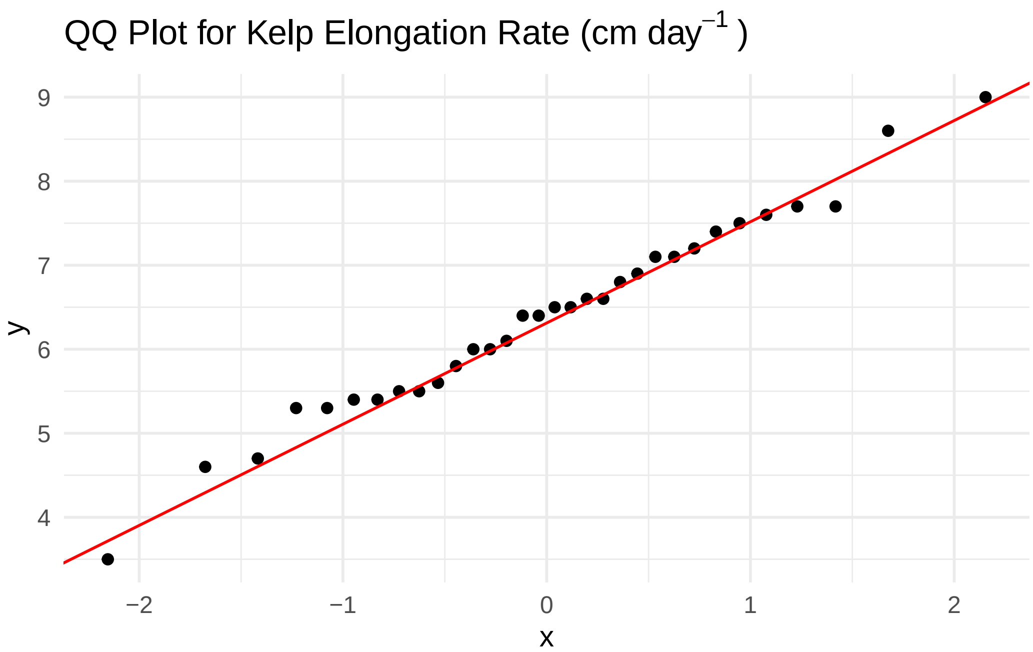
```
# Creating a ggplot base layer using kelp dataset
  ggplot(data = kelp,
         mapping = aes(x = temp_c,
                       y = kelp_elong)) +

  # First layer is a scatter plot to visualize individual observations
  geom_point(size = 2,
             alpha = 0.7,
             color = "#2E8B57",
             shape = 21) +
  # Second layer is a line of best fit using a linear regression
  geom_smooth(method = "lm",
              color = "black",
              linewidth = 1
             ) +
  # Using custom theme
  theme_minimal() +
  # Relabeling x & y axes
  labs(x = expression("Ocean Temperature (C"^o~")"),
```

```
       y = expression("Kelp Elongation Rate (cm day"^-1~")"),
       title = "The Effect of Ocean Water Temperature On
       Kelp Elongation Rate")
```

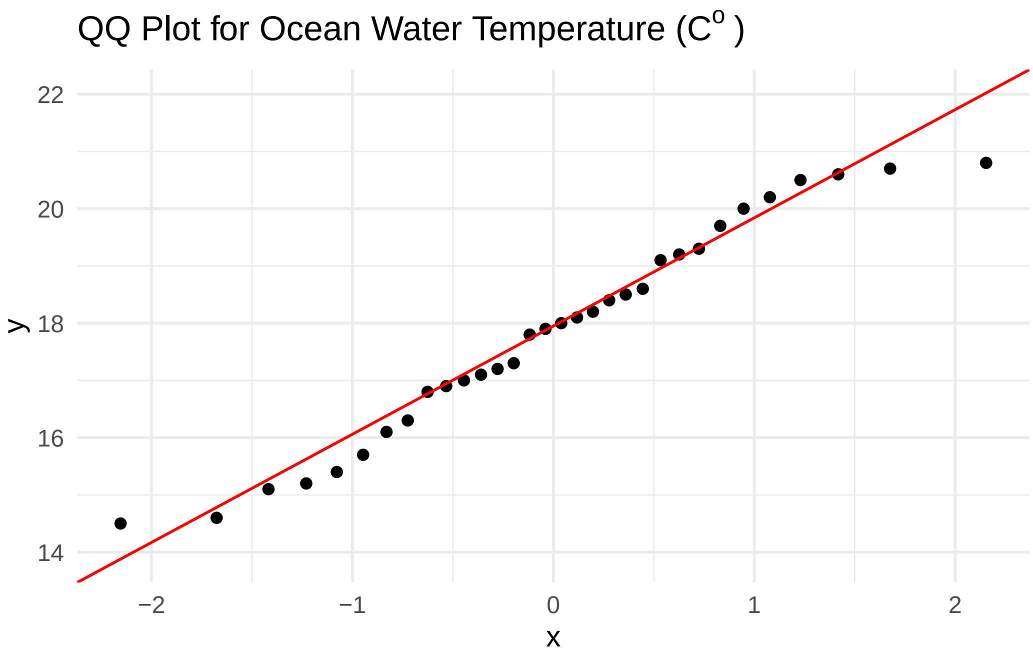
```
`geom_smooth()` using formula = 'y ~ x'
```




```
# ggplot as base layer
  ggplot(data = kelp,
         mapping = aes(sample = kelp_elong)) +
  # first layer is QQ plot
  geom_qq() +
  # second layer is QQ line
  geom_qq_line(color = "red") +
  # updating theme
  theme_minimal() +
  # updating title
  labs(title = expression("QQ Plot for Kelp Elongation Rate (cm day"^-1~")"))
```




```
# ggplot as the base layer
  ggplot(data = kelp,
   mapping = aes(sample = temp_c)) +
  # first layer is the qq plot
  geom_qq() +
  # second layer is the qq plot line
  geom_qq_line(color = "red") +
  # updating theme
  theme_minimal() +
  # updating title
  labs(title = expression("QQ Plot for Ocean Water Temperature (C"^o~")"))
```



Write-Up:

To evaluate whether a Pearson correlation test or a Spearman-rank correlation test should be used, I assessed the linearity of the two data in addition to each variable's distribution. Using a scatterplot with a fitted linear regression, it was determined that the data *appears* to have somewhat of a negatively linear relationship where as ocean temperature increases, kelp elongation rate decreases. Using qq plots, it was determined that both Ocean Temperature and Kelp Elongation Rate are normally distributed as data points fall along a liner line of best fit.

```
# Running a Pearson corrleation test
cor.test(x = kelp$temp_c,
         y = kelp$kelp_elong,
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
```

```
sample estimates:
       cor
-0.6877489
```

d. Results Communication

To evaluate the strength of the relationship between ocean temperature and giant kelp frond elongation rate, I used a Pearson correlation test as both variables were continuous and quantitative, followed a relatively negatively linear relationship, and had a normal distribution. In running the test, a strong negative relationship was found between ocean temperature and giant kelp frond elongation rate where kelp elongation rate decreases as ocean temperature increases (Pearson's product-moment correlation, t(30) = -5.19, $p < 0.01$, $\alpha = 0.05$, CI [-0.836, -0.446]).

e. Test Implications

The results of this suggest that increasing ocean temperatures are associated with significantly lower giant kelp front elongation rates. As giant kelp forests provide habitat, food, and ecosystem stability for marine organisms, reduce growth rates could have wide-spread ecological consequences as ocean temperatures continue to rise due to climate change. ### f. Double Check Your Own Work

```r
# Running a spearman-rank correlation test
cor.test(x = kelp$temp_c,
         y = kelp$kelp_elong,
         method = "spearman")
```

```
Warning in cor.test.default(x = kelp$temp_c, y = kelp$kelp_elong, method =
"spearman"): Cannot compute exact p-value with ties


    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

Write-Up:

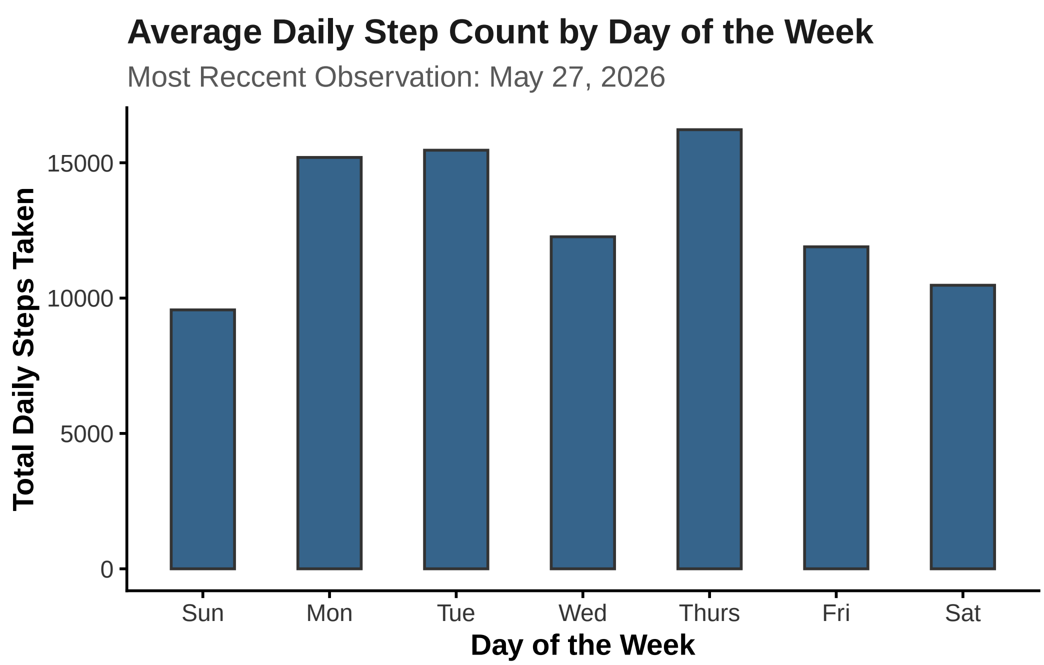
Both the Pearson correlation test ($p < 0.01$) and the Spearman-rank correlation test ($p < 0.01$) both gave results that rejected the null hypothesis of no relationship existing between ocean temperature and giant kelp frond elongation rates. For both tests, the p-values were less than $\alpha = 0.05$. Additionally, both tests indicated a strong negative relationship between the two variables with the Pearson test identifying a strong negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's product-moment correlation, $t(30) = -5.19$, $p < 0.01$, $\alpha = 0.05$, CI[-0.836, -0.446]) and the Spearman rank correlation test identifying a strong negative monotonic relationship (Spearman rank correlation test: $\rho = -0.689$, $S = 9216$, $p < 0.01$). These results indicate that as ocean temperature increases, kelp elongation rate consistently decreases.

Problem 2 - Personal Data

a. Updating Your Visualizations

```
# cleaning my data before creating ggplot
my_data_clean <- step_count |>
  # cleaning names
  clean_names() |>
  # removing any NA values
  filter(!is.na(day_of_the_week))

# creating summary data to find the mean steps per each day of the week
my_data_summary <- my_data_clean |>
  # grouping the days of the week
  group_by(day_of_the_week) |>
  # calculating the mean number of steps for each day of the week
  summarise(mean_steps = mean(total_steps_taken_for_the_day)) |>
  # ungrouping the data for good measure
  ungroup()

# ggplot as base layer
ggplot(data = my_data_summary,
       # designating x and y axis to specified columns within the dataset
       mapping = aes(x = fct_relevel(day_of_the_week,
                            "Sun",
                            "Mon",
                            "Tue",
                            "Wed",
```

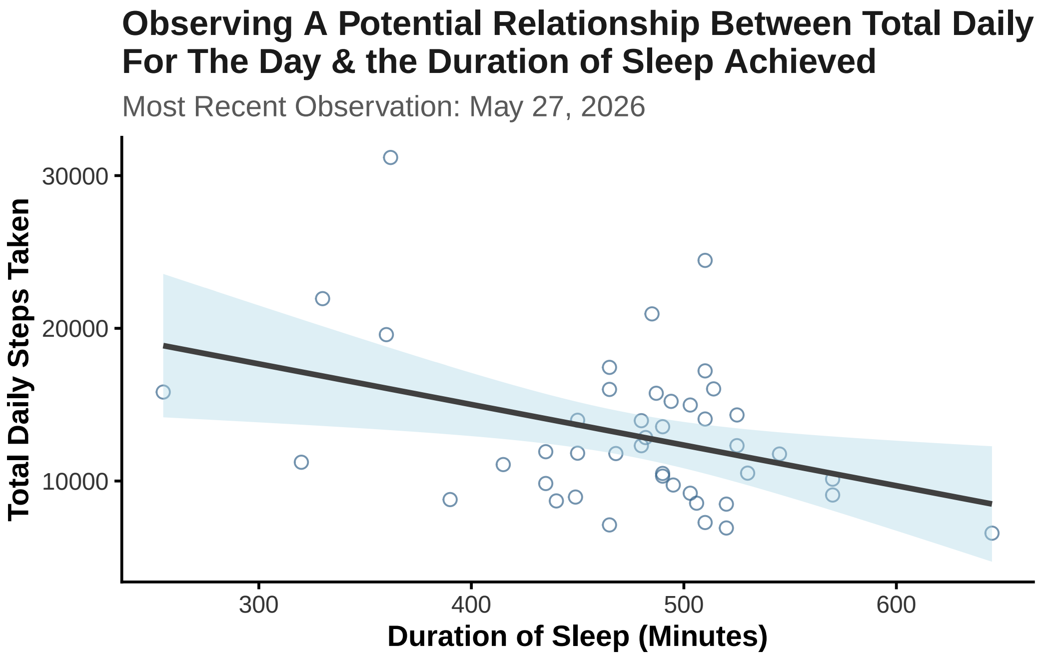
```r
                        "Thurs",
                        "Fri",
                        "Sat"), y = mean_steps)) +
  # Adding first layer
  geom_col(fill = "steelblue4",
           color = "gray20",
           width = 0.5) +
  labs(x = "Day of the Week",
       y = "Total Daily Steps Taken",
       title = "Average Daily Step Count by Day of the Week",
       subtitle = "Most Reccent Observation: May 27, 2026") +
  theme_classic() +
  # Customizing plot text and appearance
  theme(plot.title = element_text(face = "bold",
                                  color = "gray10"),
        plot.subtitle = element_text(color = "gray35"),
        axis.title = element_text(face = "bold"),
        axis.text = element_text(color = "gray20"))
```




```r
#creating ggplot base layer
ggplot(data = my_data_clean,
       mapping = aes(x = duration_of_sleep_minutes,
```

```r
                    y = total_steps_taken_for_the_day)) +
  # Adding a QQ plot
  geom_point(color = "steelblue4",
             size = 2,
             alpha = 0.7,
             shape = 21) +
  # Adding a linear regression trendline
  geom_smooth(method = "lm",
              color = "gray25",
              fill = "lightblue") +
  # Adding custom labels for the x and y axis
  labs(x = "Duration of Sleep (Minutes)",
       y = "Total Daily Steps Taken",
       title = "Observing A Potential Relationship Between Total Daily Steps Taken
For The Day & the Duration of Sleep Achieved",
      subtitle = "Most Recent Observation: May 27, 2026") +
  # Updating the theme to remove grid lines
  theme_classic() +
  # Customizing the theme elements
   theme(plot.title = element_text(face = "bold",
                                   color = "gray10"),
        plot.subtitle = element_text(color = "gray35"),
        axis.title = element_text(face = "bold"),
        axis.text = element_text(color = "gray20"))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



b. Captions

Figure 1:

Average daily step count grouped by day of the week using observations collected over the course of the observation period. Thursday had the highest average daily step count while Sunday had the lowest average daily step count, suggesting that activity levels varied noticeably depending on the day of the week.

Figure 2:

The relationship between sleep duration (minutes) and total daily step count across all observations collected during the study period. The fitted linear regression line suggests a slight negative relationship between sleep duration and daily activity, indicating that higher sleep durations were generally associated with lower total daily step counts.

Problem 3 - Affective Visualization

a. Describe An Affective Visualization

b. Create A Sketch

c. Make A Draft

d. Write An Artist Statement

e. Prep Your Materials

View my slides Here

Problem 4 - Statistical Critique

a. Revisit and Summarize

b. Visual Clarity

c. Aesthetic Clarity

d. Reccomendations